

A stable 1-semiadditive category that is not 2-semiadditive

Abstract

For every prime p , we construct an object in the category of p -typical 1-commutative monoids in p -local spectra whose underlying spectrum is the p -complete sphere and on which the cardinality of BC_p induces the zero map on underlying spectra. The construction assigns to a finite groupoid the group completion of its finite covers, with transfers that retain only components having trivial relative stabilizer. The Segal conjecture supplies the Segal condition after p -completion. Lee's vanishing theorem then implies that the norm of the constant B^2C_p -diagram on this object is not an equivalence. This gives a counterexample to the assertion that every presentable stable 1-semiadditive category is infinitely semiadditive. We also prove that the assertion holds for compactly generated p -local categories, using a finite-skeleton criterion for bounded stable chromatic height, and that a compact tensor unit forces rationality.

1 Introduction

In a presentable stable ∞ -category, a finite group action has a canonical norm from homotopy orbits to homotopy fixed points. A category is 1-semiadditive if these norms are equivalences for every finite group action. Higher semiadditivity extends this requirement to diagrams indexed by π -finite spaces, with the norms constructed inductively from those in lower degrees.

Carmeli, Schlank, and Yanovski proved that $T(n)$ -local spectra are ∞ -semiadditive and that, for localizations of spectra at weak rings, 1-semiadditivity implies ∞ -semiadditivity [CSY-A, Theorem 5.3.1 and Corollary 5.4.9]. They conjectured the same implication for all presentable stable ∞ -categories [CSY-A, Conjecture 1.1.5]. We construct a counterexample.

Fix a prime p . Write $\mathrm{Sp}_{(p)}$ for the category of p -local spectra and $\mathbb{S}_p^\wedge$ for the p -complete sphere. Let

$$\mathcal{U} = \mathrm{CMon}_1^{(p)}(\mathrm{Sp}_{(p)})$$

be the category of p -typical 1-commutative monoids in p -local spectra, and let $U : \mathcal{U} \rightarrow \mathrm{Sp}_{(p)}$ be evaluation at the point. Its objects are spectrum-valued functors on spans of finite p -groupoids satisfying the Segal condition; we recall this description in Section 2. For an object X of a presentable category and a space B , write $X[B]$ and X^B for the colimit and limit of the constant B -diagram on X .

Theorem 1.1. *The category $\mathcal{U}$ is presentable, stable, and 1-semiadditive. There is an object $M \in \mathcal{U}$ such that*

$$U(M) \simeq \mathbb{S}_p^\wedge, \quad U(|BC_p|_M) \simeq 0,$$

and the canonical norm

$$N_{B^2C_p, M} : M[B^2C_p] \longrightarrow M^{B^2C_p}$$

is not an equivalence. In particular, $\mathcal{U}$ is not 2-semiadditive.

Here $|BC_p|$ is the natural endomorphism of the identity furnished by 1-semiadditivity. The object M is obtained by changing the transfers on finite covers. Along $BG \rightarrow *$, the transfer sends a finite G -set to the set of its free orbits. To construct these transfers coherently, we start with all finite p -groupoids over a base and quotient out the components with nontrivial relative stabilizer. This quotient leaves the entire restriction functor on finite covers unchanged. Its Segal comparison after group completion and p -completion is therefore the comparison in the Segal conjecture used by Yuan [Yua, proof of Proposition 3.1]. For a nontrivial G , a trivial G -set has no free orbits, so the resulting map $U(|BG|_M)$ is zero.

The obstruction to 2-semiadditivity is then formal. Lee’s theorem gives $\mathbb{S}_p^\wedge \simeq (\mathbb{S}_p^\wedge)^{B^2C_p}$ [Yua, Corollary 1.4.1 and its proof]. Since U preserves limits and detects equivalences, the map $M \rightarrow M^{B^2C_p}$ is an equivalence. If the norm were also an equivalence, its basepoint matrix coefficient would force $|BC_p|_M$ to have a left inverse. Applying U would make the zero endomorphism of $\mathbb{S}_p^\wedge$ left invertible.

Compact generation gives a positive result. An object X in a presentable stable p -local category has *finite stable chromatic height* if $V \otimes X \simeq 0$ for some nonzero finite p -local spectrum V . The tensor product here denotes the canonical action of spectra.

Theorem 1.2. *Let $\mathcal{C}$ be a presentable stable p -local 1-semiadditive ∞ -category. Every compact object of $\mathcal{C}$ has finite stable chromatic height. If $\mathcal{C}$ is compactly generated, then it is ∞ -semiadditive. If $\mathcal{C}$ is presentably symmetric monoidal and its tensor unit is compact, then $\mathcal{C}$ is rational.*

The proof in Section 5 uses the inverse norm on a compact object to split the transfer on a finite skeleton of BC_p . The dimension of that skeleton bounds the Morava support of its endomorphism spectrum. The weak-ring nilpotence theorem and the bootstrap theorem of Carmeli–Schlank–Yanovski then apply. A module category built from Yuan’s height-1 sphere shows that this finite-skeleton property need not hold for noncompact objects, even in an ∞ -semiadditive category. Section 6 records the corresponding comparison between localization and norm defects.

2 Norms and spectrum-valued 1-commutative monoids

All categories and functors are understood in the ∞ -categorical sense. A space is π -*finite* if it has finitely many components, finite homotopy groups, and only finitely many nonzero homotopy groups. It is m -*finite* if it is also m -truncated. A finite p -groupoid is a finite disjoint union of classifying spaces of finite p -groups. Let Gpd_p denote their category and put

$$\mathrm{Span}_p = \mathrm{Span}(\mathrm{Gpd}_p).$$

A morphism is a span $A \leftarrow E \rightarrow B$, with composition by homotopy pullback. These pullbacks remain in Gpd_p : their isotropy groups are subgroups of products of finite p -groups, and their component sets are finite. We use $\mathcal{S}$ for spaces and $(-)^{\simeq}$ for the maximal subgroupoid of a category: it retains every object and only the equivalences between objects. The notation $\mathrm{CMon}(\mathcal{S})$ denotes commutative monoid objects in spaces, also called E_∞ -monoid spaces.

2.1 The Segal condition

A p -typical 1-commutative monoid in $\mathrm{Sp}_{(p)}$ is a functor $M : \mathbf{Span}_p \rightarrow \mathrm{Sp}_{(p)}$ for which the restrictions to points induce equivalences

$$\rho_A : M(A) \xrightarrow{\sim} M(*)^A \quad (A \in \mathbf{Gpd}_p). \quad (2.1)$$

The restrictions in this formula form an A -indexed coherent family. This is the span description of higher commutative monoids [Har, Section 5.2]; we use its p -typical version as in [Yua, Definition 2.3].

Proposition 2.1. *The category $\mathcal{U}$ is presentable, stable, p -local, and 1-semiadditive. Evaluation at the point $U : \mathcal{U} \rightarrow \mathrm{Sp}_{(p)}$ preserves all limits and is conservative.*

Proof. The universal property of higher commutative monoids gives presentability and p -typical 1-semiadditivity; specifically, [Yua, Proposition 2.7(1)–(3)] identifies the construction as tensoring presentable categories with the mode for p -typical 1-semiadditivity.

Limits of functors satisfying (2.1) are computed pointwise, since limits commute with limits. The zero functor satisfies the Segal condition, and so do pointwise fibers. Pointwise suspension and desuspension preserve the condition, because they are inverse equivalences of spectra and commute with limits. The Segal subcategory is therefore stable. Multiplication by every prime different from p is an equivalence pointwise, so it is p -local. It admits all small limits and colimits; [CSY-H, Proposition 3.2.6] upgrades its p -typical 1-semiadditivity to 1-semiadditivity.

Pointwise computation of limits proves the assertion about U . If a natural transformation is an equivalence at $*$, the Segal equivalences show that it is an equivalence at every $A \in \mathbf{Gpd}_p$. Thus U is conservative. $\square$

For $A \in \mathbf{Gpd}_p$, the cardinality endomorphism $|A|_M$ is induced, on evaluation at $*$, by the span

$$* \longleftarrow A \longrightarrow *. \quad (2.2)$$

This description agrees with the cardinality supplied by semiadditivity [Yua, Construction 2.4 and Proposition 2.7(4)].

2.2 A matrix coefficient of the norm

Let $\mathcal{C}$ be a presentable 1-semiadditive category. For a connected pointed 2-finite space B and $X \in \mathcal{C}$, write

$$i : X \longrightarrow X[B], \quad e : X^B \longrightarrow X, \quad \nabla : X[B] \longrightarrow X$$

for the basepoint inclusion, evaluation, and collapse maps. The constant cone gives $\Delta : X \rightarrow X^B$, and

$$e\Delta = \mathrm{id}_X, \quad \nabla i = \mathrm{id}_X. \quad (2.3)$$

The norm $N_{B,X} : X[B] \rightarrow X^B$ exists because the diagonal of B has 1-finite fibers. Its basepoint coefficient is

$$eN_{B,X}i = |\Omega B|_X. \quad (2.4)$$

Indeed, the canonical norm is defined by integrating the identity along the diagonal [CSY-A, Definition 3.1.3 and Remark 3.1.4]. The pullback of the diagonal to the pair of basepoints is $* \times_B * \simeq \Omega B$. Its integral is the right side of (2.4).

Lemma 2.2. *Suppose that $\Delta : X \rightarrow X^B$ and $N_{B,X} : X[B] \rightarrow X^B$ are equivalences. Then $|\Omega B|_X$ has a left inverse.*

Proof. By (2.3), e is inverse to Δ . Thus (2.4) gives $N_{B,X}i = \Delta|\Omega B|_X$. Setting $s = N_{B,X}^{-1}\Delta$ yields $i = s|\Omega B|_X$. Composing with ∇ proves

$$(\nabla s)|\Omega B|_X = \text{id}_X. \quad \square$$

3 Transfers on finite covers

For $A \in \mathbf{Gpd}_p$, a finite cover of A is a map with finite discrete homotopy fibers. Finite covers and their equivalences form the space $(\mathbf{Fin}^A)^\simeq$, with disjoint union as its E_∞ -monoid operation. We construct a functor with these values whose transfers select the components that remain finite covers.

Proposition 3.1. *There is a functor*

$$\mathbf{Cov} : \mathbf{Span}_p \longrightarrow \mathbf{CMon}(\mathcal{S})$$

*whose restriction to $\mathbf{Gpd}_p^{\text{op}}$ is naturally equivalent to $A \mapsto (\mathbf{Fin}^A)^\simeq$, with the usual pullback maps on finite covers. For a span $A \leftarrow E \rightarrow B$, its value on a cover $X \rightarrow A$ is the union of those connected components of $X \times_A E \rightarrow B$ that are finite covers. In particular, the map induced by $BG \rightarrow *$ sends a finite G -set to X_{free}/G .*

The construction begins with all finite groupoids over a base. Components with nontrivial relative stabilizer are preserved by every span; quotienting them out will give the functor in Proposition 3.1.

3.1 Relative stabilizers

Let $\mathbf{Fam}(A)$ be the space of finite p -groupoids over A . Thus a point of $\mathbf{Fam}(A)$ is a map $f : X \rightarrow A$, and its paths are equivalences over A . In slice-category notation,

$$\mathbf{Fam}(A) = (\mathbf{Gpd}_{p/A})^\simeq.$$

Here the subscript $/A$ means that the objects are maps to the fixed base A ; the superscript $\simeq$ keeps only equivalences between these maps. Disjoint union equips it with an E_∞ -monoid structure. The span $A \xleftarrow{u} E \xrightarrow{v} B$ sends $f : X \rightarrow A$ to

$$X \times_A E \longrightarrow E \xrightarrow{v} B. \quad (3.1)$$

This defines a functor $\mathbf{Fam} : \mathbf{Span}_p \rightarrow \mathbf{CMon}(\mathcal{S})$. On underlying spaces it is the covariant representable functor $\mathbf{Map}_{\mathbf{Span}_p}(*, -)$; composition of spans distributes over disjoint union, giving its indicated additive refinement.

For $f : X \rightarrow A$, the *relative stabilizer* of $x \in X$ is the group of automorphisms of x that induce the identity of $f(x)$ in A . We write

$$\text{Stab}_f(x) = \ker(\text{Aut}_X(x) \xrightarrow{f} \text{Aut}_A(f(x))).$$

Whether $\text{Stab}_f(x)$ is trivial is constant on each connected component of X , and invariant under equivalences over A . Let $\text{Discard}(A) \subseteq \text{Fam}(A)$ consist of the objects all of whose connected components have nontrivial relative stabilizer. The name *Discard* records the part to be removed from *Fam*. The condition is imposed on every component, so the empty groupoid is included. For example, $BC_p \rightarrow *$ belongs to $\text{Discard}(*)$, whereas $* \sqcup BC_p \rightarrow *$ does not, because its point component has trivial stabilizer.

Lemma 3.2. *The inclusions $\text{Discard}(A) \subseteq \text{Fam}(A)$ define a subfunctor $\text{Discard} \rightarrow \text{Fam}$ with values in $\text{CMon}(\mathcal{S})$.*

Proof. Take $f : X \rightarrow A$ in $\text{Discard}(A)$ and a span $A \xleftarrow{u} E \xrightarrow{v} B$. At an object $y = (x, e, \theta)$ of $Y = X \times_A E$, where $\theta : f(x) \simeq u(e)$, there is an injection

$$\begin{aligned} \text{Stab}_f(x) &\hookrightarrow \ker(\text{Aut}_Y(y) \longrightarrow \text{Aut}_B(v(e))), \\ h &\longmapsto (h, 1_e). \end{aligned} \tag{3.2}$$

The pair satisfies the pullback compatibility equation because $f(h) = 1$, and its image in B is the identity. Its first component recovers h , so the map is injective. The target kernel is therefore nontrivial at every object of Y .

The specified subspaces $\text{Discard}(A)$ are unions of connected components of $\text{Fam}(A)$. The preceding calculation shows that every structure map of *Fam* preserves them. Consequently all composition equivalences and higher coherences restrict to these subspaces. Disjoint unions also preserve the condition, giving the claimed subfunctor of commutative monoids. $\square$

3.2 The quotient and its restrictions

Form the pushout

$$\text{Cov} = \text{Fam} \amalg_{\text{Discard}0} \quad \text{in} \quad \text{Fun}(\text{Span}_p, \text{CMon}(\mathcal{S})). \tag{3.3}$$

The symbol 0 denotes the trivial commutative monoid. Colimits in this functor category are computed pointwise.

Lemma 3.3. *For every $A \in \text{Gpd}_p$, disjoint union gives a decomposition*

$$\text{Fam}(A) \simeq (\text{Fin}^A)^{\simeq} \oplus \text{Discard}(A) \quad \text{in} \quad \text{CMon}(\mathcal{S}).$$

These decompositions are natural under restriction along maps of groupoids. In particular,

$$\text{Cov}|_{\text{Gpd}_p^{\text{op}}} \simeq (A \longmapsto (\text{Fin}^A)^{\simeq}) \quad \text{in} \quad \text{Fun}(\text{Gpd}_p^{\text{op}}, \text{CMon}(\mathcal{S})). \tag{3.4}$$

Proof. A map $f : X \rightarrow A$ is a finite cover if and only if $\text{Stab}_f(x)$ is trivial for every x . Indeed, its homotopy fibers are finite groupoids, and their automorphism groups are precisely these kernels. A groupoid with trivial automorphism groups is equivalent to a discrete space.

Let $\text{Conn}(A)$ be the moduli space of connected objects of Gpd_p over A . Unique decomposition into connected components gives

$$\text{Fam}(A) \simeq \text{Sym}(\text{Conn}(A)),$$

where *Sym* is the free E_∞ -monoid functor. Partition $\text{Conn}(A)$ into the subspaces where the relative stabilizer is trivial and nontrivial. The two free monoids are respectively $(\text{Fin}^A)^{\simeq}$ and

$\text{Discard}(A)$, and the free monoid functor carries their disjoint union to their coproduct. This proves the asserted decomposition, including its automorphism spaces.

For a pure restriction along $B \rightarrow A$, the homotopy fiber of $X \times_A B \rightarrow B$ over b is the homotopy fiber of $X \rightarrow A$ over its image. Thus pullback preserves both classes of components. Selecting the two unions of components commutes with these pullbacks and their natural equivalences, so the decomposition is natural on Gpd_p^{op} . Taking the pushout in (3.3) gives

$$((\text{Fin}^A)^\simeq \oplus \text{Discard}(A)) \amalg_{\text{Discard}(A)} 0 \simeq (\text{Fin}^A)^\simeq$$

naturally under restriction. This is (3.4). $\square$

Proof of Proposition 3.1. Use the functor (3.3). Lemma 3.3 identifies its restriction with the usual pullback functor on finite covers. The map from $\text{Fam}(A)$ to this quotient keeps the components with trivial relative stabilizer. Applying it to (3.1) gives the stated span action.

A cover of BG is the action groupoid $X//G \rightarrow BG$ of a finite G -set X . The automorphism group of x in $X//G$ is $\text{Stab}_G(x)$. After composition with $BG \rightarrow *$, its relative stabilizer is therefore trivial exactly on the free orbits. Each such component is contractible, and their space is X_{free}/G . $\square$

Remark 3.4. For a homomorphism $\varphi : G \rightarrow H$ of finite p -groups, the transfer of a finite G -set has the concrete description

$$X \longmapsto H \times_G \{x \in X : \text{Stab}_G(x) \cap \ker \varphi = 1\}.$$

It acts on the groupoids of finite sets and equivariant isomorphisms. For a G -orbit G/S and a second homomorphism $\psi : H \rightarrow K$, survival under the two successive transfers is equivalent to

$$S \cap \ker \varphi = 1, \quad \varphi(S) \cap \ker \psi = 1,$$

or, equivalently, $S \cap \ker(\psi\varphi) = 1$.

4 The spectral object and its second norm

For an E_∞ -monoid space T , let T^{gp} denote its group completion, regarded as a connective spectrum via $\text{CGrp}(\mathcal{S}) \simeq \text{Sp}_{\geq 0}$. Applying this construction and then p -completion pointwise to (3.3) gives

$$M(A) = (\text{Cov}(A)^{\text{gp}})_p^\wedge. \quad (4.1)$$

For a category with disjoint unions, write $K_\oplus$ for the spectrum obtained by group completing its maximal subgroupoid under disjoint union. Lemma 3.3 gives a natural equivalence on restrictions

$$M|_{\text{Gpd}_p^{op}} \simeq (A \longmapsto K_\oplus(\text{Fin}^A)_p^\wedge). \quad (4.2)$$

Proposition 4.1. *The functor M belongs to $\mathcal{U}$. Its value at the point is $\mathbb{S}_p^\wedge$, and for every nontrivial finite p -group G ,*

$$U(|BG|_M) \simeq 0.$$

Proof. The Barratt–Priddy–Quillen equivalence identifies $K_{\oplus}(\text{Fin})$ with $\mathbb{S}$, and hence $M(*) \simeq \mathbb{S}_p^{\wedge}$; see also the first assertion in the proof of [Yua, Proposition 3.1].

The remaining Segal comparison is determined by the restriction functor in (4.2). For $A = BG$ it is

$$K_{\oplus}(\text{Fin}^{BG})_p^{\wedge} \longrightarrow (\mathbb{S}_p^{\wedge})^{BG}. \quad (4.3)$$

For a finite p -group G , this is the p -completed Segal comparison, which is an equivalence by Carlsson’s theorem in the form used in [Yua, proof of Proposition 3.1]. Both sides take finite disjoint unions of groupoids to products. For the left side, this follows from the product decomposition of finite covers, the preservation of finite products by group completion, and exactness of p -completion on spectra. Every object of Gpd_p is such a union of classifying spaces, so (2.1) holds for every A . Thus $M \in \mathcal{U}$.

Under Proposition 3.1, the span (2.2) for $A = BG$ acts by

$$\text{Fin}^{\simeq} \xrightarrow{\text{triv}} (\text{Fin}^{BG})^{\simeq} \xrightarrow{(-)_{\text{free}}/G} \text{Fin}^{\simeq}.$$

For $G \neq 1$, the composite is the constant functor with value the empty set. It is the zero map of additive E_{∞} -monoids. Group completion and p -completion therefore yield the zero map of spectra, as claimed. $\square$

Remark 4.2. Yuan’s functor has the same restriction functor (4.2), but its transfer along $BG \rightarrow *$ takes all orbits [Yua, Proposition 3.1 and its proof]. On a trivial G -set this transfer returns the original set, so its cardinality map on the underlying sphere is the identity. The quotient construction of Section 3 changes that transfer to free orbits and gives the zero cardinality map in Proposition 4.1.

Proof of Theorem 1.1. Proposition 2.1 gives the assertions about $\mathcal{U}$, and Proposition 4.1 constructs the required object. It remains to test its B^2C_p -norm.

The space B^2C_p is simply connected and π -finite. Lee’s theorem, in the p -complete form established in the proof of [Yua, Corollary 1.4.1], gives the equivalence

$$\mathbb{S}_p^{\wedge} \xrightarrow{\simeq} (\mathbb{S}_p^{\wedge})^{B^2C_p}$$

induced by $B^2C_p \rightarrow *$. Since U preserves limits, this is the underlying map of $\Delta_M : M \rightarrow M^{B^2C_p}$. Since U is conservative, Δ_M is an equivalence.

If $N_{B^2C_p, M}$ were an equivalence, Lemma 2.2 would give an endomorphism b of M with

$$b|_{BC_p|_M} = \text{id}_M.$$

Applying U and using Proposition 4.1 would give

$$U(b) \circ 0 = \text{id}_{\mathbb{S}_p^{\wedge}}.$$

This contradicts $\pi_0(\mathbb{S}_p^{\wedge}) = \mathbb{Z}_p \neq 0$. Thus the norm is not an equivalence. $\square$

5 Compact objects and finite stable chromatic height

Throughout this section, $\mathcal{C}$ is a presentable stable p -local 1-semiadditive ∞ -category. We write $\mathrm{map}_{\mathcal{C}}(X, Y)$ for its mapping spectrum and $E \otimes X$ for its tensoring over p -local spectra.

Choose the bar construction model of BC_p , with one vertex and finitely many cells in each dimension. Let B_d be its d -skeleton, and set

$$F_d = (\Sigma_+^\infty B_d)_{(p)}.$$

The stable transfer for the universal C_p -cover is a map

$$\tau : (\Sigma_+^\infty BC_p)_{(p)} \longrightarrow \mathbb{S}_{(p)}.$$

Its restriction to F_d will be denoted by τ_d . Restriction along the vertex gives the degree- p endomorphism of $\mathbb{S}_{(p)}$.

Lemma 5.1. *For every $d \geq 0$, the map $H\mathbb{F}_p \wedge \tau_d$ is nullhomotopic. For $n \geq 1$, the map $K(n) \wedge \tau_d$ is nullhomotopic whenever*

$$d < 2(p^n - 1).$$

Proof. The free-module adjunction identifies the homotopy class of $K(n) \wedge \tau_d$ with the class

$$[\eta_{K(n)} \circ \tau_d] \in K(n)^0(B_d),$$

where $\eta_{K(n)} : \mathbb{S}_{(p)} \rightarrow K(n)$ is the unit. We show that restriction to the vertex is an isomorphism on $K(n)^0$ in the stated range.

Write $D = 2(p^n - 1)$. The skeletal filtration of the finite CW complex B_d gives the cohomological Atiyah–Hirzebruch spectral sequence

$$E_2^{s,t} = H^s(B_d; K(n)^t(*)) \implies K(n)^{s+t}(B_d), \quad d_r : E_r^{s,t} \longrightarrow E_r^{s+r, t-r+1}.$$

Here $0 \leq s \leq d$, and the coefficient group $K(n)^t(*)$ vanishes unless t is a multiple of D . The filtration is finite, so the spectral sequence converges with a finite, exhaustive, and separated filtration on each abutment group.

If $s + t = 0$ and $d < D$, the only possibly nonzero term is $E_2^{0,0} = H^0(B_d; \mathbb{F}_p) = \mathbb{F}_p$. There are no incoming differentials to $(0, 0)$. An outgoing differential has target $(r, 1 - r)$. For $2 \leq r \leq d < D$, its coefficient degree $1 - r$ is not a multiple of D ; for $r > d$, its target lies outside the filtration. Thus this term survives, and it is the only nonzero associated graded piece of total degree zero. There are consequently no extension choices in that degree. Naturality for the inclusion of the vertex identifies the resulting isomorphism with

$$K(n)^0(B_d) \xrightarrow{\simeq} K(n)^0(*).$$

The transfer class restricts to $p \cdot 1 = 0$, so it vanishes. The free-module adjunction then gives the asserted nullhomotopy.

For $H\mathbb{F}_p$, the same conclusion follows from $H^0(B_d; \mathbb{F}_p) \simeq H^0(*; \mathbb{F}_p)$, since B_d is connected and the restricted transfer again has degree p . $\square$

The norm equivalence for the constant BC_p -diagram defines a map

$$s_X : X \xrightarrow{\Delta} X^{BC_p} \xrightarrow{N_{BC_p, X}^{-1}} X[BC_p]. \quad (5.1)$$

The evaluation map $\text{ev} : X^{BC_p} \rightarrow X$ satisfies $\text{ev} \circ \Delta = \text{id}_X$. The norm construction also gives

$$\text{ev} \circ N_{BC_p, X} = \tau \otimes X. \quad (5.2)$$

Indeed, evaluation at the vertex identifies the norm with summation over the fibers of the universal C_p -cover; this is the stable transfer tensored with X . This is the transfer description of the norm recalled in [CSY-A, Section 1.2]. In particular, $(\tau \otimes X) \circ s_X = \text{id}_X$.

Proposition 5.2. *Suppose that s_X factors through $F_d \otimes X$ for some $d \geq 0$. Then*

$$H\mathbb{F}_p \wedge \text{map}_{\mathcal{C}}(X, X) = 0,$$

and, for every $n \geq 1$ with $d < 2(p^n - 1)$,

$$K(n) \wedge \text{map}_{\mathcal{C}}(X, X) = 0.$$

Moreover, X has finite stable chromatic height.

Proof. Let $u : X \rightarrow F_d \otimes X$ be a factorization of s_X . Equation (5.2) gives

$$(\tau_d \otimes X) \circ u = \text{id}_X. \quad (5.3)$$

We first translate this identity into an identity involving the unit of the endomorphism ring spectrum. The transfer calculation will then bound its chromatic support, and a finite spectrum annihilating this ring will also annihilate X .

In this proof, let $R = \text{map}_{\mathcal{C}}(X, X)$, with its composition multiplication and unit $\eta_R : \mathbb{S}_{(p)} \rightarrow R$. For every finite p -local spectrum E , the canonical map

$$E \wedge R \longrightarrow \text{map}_{\mathcal{C}}(X, E \otimes X)$$

is an equivalence. It is an equivalence for $E = \mathbb{S}_{(p)}$; both sides are exact in E and preserve retracts, so it is an equivalence for every finite E . Applying this to F_d , the map u corresponds to a map $b : \mathbb{S}_{(p)} \rightarrow F_d \wedge R$ such that

$$(\tau_d \wedge R) \circ b = \eta_R. \quad (5.4)$$

Let E be $H\mathbb{F}_p$, or let $E = K(n)$ in the range of the proposition. Lemma 5.1 and (5.4) imply that $E \wedge \eta_R$ is nullhomotopic. The composite

$$E \wedge R \xrightarrow{E \wedge \eta_R \wedge R} E \wedge R \wedge R \xrightarrow{E \wedge \mu_R} E \wedge R$$

is the identity by the unit axiom, and it is nullhomotopic by the preceding observation. Hence $E \wedge R = 0$, proving both vanishing assertions.

If $X = 0$, the final assertion holds with $V = \mathbb{S}_{(p)}$. Otherwise R is a nonzero p -local associative ring spectrum, and therefore a weak ring in the sense of [CSY-A]. Choose an integer $N \geq 0$ such that $d < 2(p^n - 1)$ for every $n > N$. We have proved that $K(n) \wedge R = 0$ for $n > N$ and that

$H\mathbb{F}_p \wedge R = 0$. Thus its chromatic support, including the height- ∞ theory $H\mathbb{F}_p$, is contained in $\{0, \dots, N\}$. By [CSY-A, Theorem 5.4.4], there is a nonzero finite p -local spectrum V of type $N + 1$ such that $V \wedge R = 0$. Finite duality and the preceding mapping-spectrum identity give

$$\mathrm{map}_{\mathcal{C}}(V \otimes X, V \otimes X) \simeq DV \wedge V \wedge R = 0.$$

The identity of $V \otimes X$ is therefore nullhomotopic, so $V \otimes X = 0$. $\square$

Theorem 5.3. *Every compact object of $\mathcal{C}$ has finite stable chromatic height. If $\mathcal{C}$ is compactly generated, then it is ∞ -semiadditive.*

Proof. For a compact object X , the map (5.1) factors through one term of the filtered colimit

$$X[BC_p] \simeq \mathrm{colim}_d (F_d \otimes X).$$

Here factorization of its homotopy class suffices for (5.3). Proposition 5.2 therefore proves the first assertion.

Assume that $\mathcal{C}$ is compactly generated. Let $\mathcal{C}_{\infty^{\mathrm{st}}}$ be the full subcategory of objects Y for which $\mathrm{Map}_{\mathcal{C}}(Z, Y)$ is contractible for every object Z of finite stable chromatic height. If $Y \in \mathcal{C}_{\infty^{\mathrm{st}}}$, the first assertion shows that every compact generator, and every shift of such a generator, has contractible mapping space into Y . The generating property implies $Y = 0$. Thus $\mathcal{C}_{\infty^{\mathrm{st}}} = 0$. The bootstrap theorem [CSY-H, Theorem 5.5.17] applies to this presentable stable p -local 1-semiadditive category and proves that $\mathcal{C}$ is ∞ -semiadditive. $\square$

5.1 A compact tensor unit

Proposition 5.4. *Suppose that $\mathcal{C}$ has a symmetric monoidal structure whose tensor product preserves colimits separately in each variable. If its tensor unit $\mathbf{1}$ is compact, then $\mathcal{C}$ is rational: every nonzero integer acts invertibly on every object.*

Proof. Compactness gives a finite-skeleton factorization of $s_{\mathbf{1}}$, and hence a section

$$u : \mathbf{1} \longrightarrow F_d \otimes \mathbf{1}, \quad (\tau_d \otimes \mathbf{1})u = \mathrm{id}_{\mathbf{1}}.$$

The vertex splits off the sphere summand, so

$$F_d \simeq \mathbb{S}_{(p)} \vee \overline{F}_d, \quad \tau_d = (p, \overline{\tau}_d),$$

where $\overline{F}_d$ is the reduced suspension spectrum of the connected based finite complex B_d , localized at p . Its degree-zero rational cohomology vanishes. Since $\overline{F}_d$ is finite, this gives

$$[\overline{F}_d, \mathbb{S}_{(p)}] \otimes \mathbb{Q} \simeq [\overline{F}_d, H\mathbb{Q}] = 0.$$

Consequently the class of $\overline{\tau}_d$ is p -primary torsion: there is a $k \geq 0$ with $p^k \overline{\tau}_d = 0$.

Write $u = (r, \overline{u})$ according to the splitting. In the commutative ring $\pi_0 \mathrm{map}_{\mathcal{C}}(\mathbf{1}, \mathbf{1})$, the section identity becomes

$$1 = pr + t, \quad t = (\overline{\tau}_d \otimes \mathbf{1})\overline{u}, \quad p^k t = 0.$$

The tensor product preserves BC_p -colimits in each variable, so $\mathcal{C}$ is 1-semiadditively symmetric monoidal. Corollary 4.3.5 of [CSY-A] applies to its unit, viewed as both a commutative algebra and a commutative coalgebra. It states that every torsion element of this endomorphism ring is nilpotent. Thus t is nilpotent, $1 - t = pr$ is invertible, and p is invertible. Tensoring its inverse on $\mathbf{1}$ with any object proves that p acts invertibly throughout $\mathcal{C}$. All integers prime to p already act invertibly because $\mathcal{C}$ is p -local. $\square$

5.2 A unit without a finite-skeleton factorization

Let $\mathcal{Y}_1$ be the universal presentable stable p -local ∞ -semiadditive category of semiadditive height 1 [CSY-H, Theorem F]. It can be identified with the full subcategory of $\mathcal{U}$ consisting of p -complete objects on which $|BC_p|$ is invertible [Yua, Proposition 2.12 and Example 2.14]. Denote the canonical adjunction by

$$L_{\mathcal{Y}_1} : \mathrm{Sp}_{(p)} \rightleftarrows \mathcal{Y}_1 : U_{\mathcal{Y}_1},$$

where $L_{\mathcal{Y}_1}$ is symmetric monoidal. Yuan constructs a commutative algebra $A \in \mathrm{CAlg}(\mathcal{Y}_1)$ with

$$U_{\mathcal{Y}_1}(A) \simeq \mathbb{S}_p^\wedge \quad (5.5)$$

as commutative ring spectra [Yua, Theorem 1.4].

Proposition 5.5. *Put $\mathcal{D} = \mathrm{Mod}_A(\mathcal{Y}_1)$. This category is presentable, stable, p -local, and ∞ -semiadditive. For its tensor unit A , the map*

$$s_A : A \longrightarrow A[BC_p]$$

does not factor through $F_d \otimes A$ for any finite d . In particular, A is not compact in $\mathcal{D}$.

Proof. The module category is presentable, stable, and p -local. Its forgetful functor to $\mathcal{Y}_1$ is conservative and creates limits and colimits, since the monad $A \otimes (-)$ preserves colimits. It therefore preserves the canonical norms. Applying this functor to a norm indexed by a π -finite space gives the corresponding norm in $\mathcal{Y}_1$, which is an equivalence. Thus $\mathcal{D}$ is ∞ -semiadditive.

The free-module adjunction and the adjunction $L_{\mathcal{Y}_1} \dashv U_{\mathcal{Y}_1}$ identify the endomorphism spectrum of the unit as

$$\mathrm{map}_{\mathcal{D}}(A, A) \simeq \mathrm{map}_{\mathcal{Y}_1}(\mathbf{1}, A) \simeq U_{\mathcal{Y}_1}(A) \simeq \mathbb{S}_p^\wedge.$$

Here the second equivalence uses $L_{\mathcal{Y}_1}(\mathbb{S}_{(p)}) = \mathbf{1}$. The p -completion map of the sphere is a mod- p homology equivalence, so

$$H\mathbb{F}_p \wedge \mathrm{map}_{\mathcal{D}}(A, A) \simeq H\mathbb{F}_p \neq 0.$$

Proposition 5.2 therefore excludes every finite-skeleton factorization of s_A . Such a factorization would exist if A were compact. $\square$

6 Localization and norm defects

The failure of a higher norm can also be compared with the failure of localization to preserve its target limit. We record the comparison for the localization at $|BC_p|$.

The category $\mathcal{U}$ has its canonical presentably symmetric monoidal structure. Write $\mathbf{1}$ for its tensor unit and set

$$a = |BC_p|_{\mathbf{1}} \in \pi_0 \mathrm{End}_{\mathcal{U}}(\mathbf{1}).$$

The action of this scalar on each object is its cardinality endomorphism. Let

$$L : \mathcal{U} \longrightarrow \mathcal{U}[a^{-1}]$$

be the localization inverting this action. Its underlying endofunctor is the telescope $X \mapsto \mathrm{colim}(X \xrightarrow{a} X \xrightarrow{a} \cdots)$. Its target is again presentable, stable, and 1-semiadditive: it is a module

category over $\mathcal{U}$ obtained by inverting an endomorphism of the tensor unit. It has semiadditive height at most 1, because $|BC_p|$ is invertible. The height bootstrap [CSY-H, Theorem 3.2.7] therefore makes the target ∞ -semiadditive.

For $F : B^2C_p \rightarrow \mathcal{U}$, define

$$T_2(F) = \text{cofib} \left(\text{colim}_{B^2C_p} F \xrightarrow{N_F} \lim_{B^2C_p} F \right)$$

and let

$$c_F : L \left(\lim_{B^2C_p} F \right) \longrightarrow \lim_{B^2C_p} LF$$

be the canonical comparison map.

Proposition 6.1. *There is a natural equivalence*

$$L(T_2(F)) \simeq \text{fib}(c_F).$$

Proof. The functor L preserves colimits and is 1-semiadditive. Compatibility of canonical norms with such functors gives, after identifying the source by preservation of colimits,

$$c_F \circ L(N_F) = N_{LF}. \tag{6.1}$$

This compatibility follows from the diagonal construction of the norm: the diagonal fibers are 1-finite and L preserves their integrals. More precisely, apply [CSY-A, Theorem 3.2.3, Proposition 3.2.2, and Definition 2.2.9] to the colimit comparison and the limit comparison c_F . The right side of (6.1) is an equivalence in the target category.

In a stable category, if a composite $A \xrightarrow{u} B \xrightarrow{v} C$ is an equivalence, it identifies u with a section of v , and its complementary summand is both $\text{cofib}(u)$ and $\text{fib}(v)$. Applying this to (6.1) and using exactness of L gives the claim. $\square$

The comparison in Proposition 6.1 identifies the localized norm defect without asserting that it vanishes. In the telescope description, the map c_F is the comparison between the sequential colimit of the B^2C_p -limits and the B^2C_p -limit of the pointwise sequential colimit.

Numbered citations refer to the arXiv versions listed below.

References

- [CSY-A] S. Carmeli, T. M. Schlank, and L. Yanovski, *Ambidexterity in chromatic homotopy theory*. [arXiv:1811.02057v3](#), 2020.
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